

Sarah CHOUCHENE
chouchene@unistra.fr
+33 771 786 710

Linkedin
https://www.linkedin.com/in/sarah-chouchene-050195

ORCID
https://orcid.org/0009-0007-7750-8645

Address
Rue François Noblat,
67000, Strasbourg
France

Nationality
French

Languages

- French- Very well
- English- Advanced
- German- Elementary A1+

Sarah CHOUCHENE Ph.D

Machine Vision • Deep Learning • 3D Reconstruction • Multidisciplinary R&D

Profile

Ph.D. engineer working at the intersection of imaging, machine learning, and physical systems. Interested in multidisciplinary projects (healthcare, industry, research) that apply machine vision, deep learning, and 3D reconstruction, with an emphasis on reproducible pipelines (MLflow) and production-minded practices (CI/CD).

Education

2020 - 2024, University of Lorraine, Institut Jean Lamour
PhD student in Plasma Physics and Machine Vision

2017 - 2020, Polytech Nancy (engineering school)
Systems control and supervision

2019 - 2020, University of Lorraine, Science and technology
Master in complex systems engineering

2016 - 2017, National Polytechnic school, Oran
Electrical Electrotechnical and Automatic engineering

Experiences

Mars 2025 - Today, ***ICube, Plateforme GAIA, Université de Strasbourg, CNRS, Strasbourg***
Research engineer:
- Developed a deep-learning method PINNs based to estimate position and orientation of magnetic sensors from measurement data; implemented training/inference workflows in Python/PyTorch with CUDA GPU acceleration, tracked experiments/metrics, versioned artifacts, and logged/shareable models with MLflow (model packaging + registry).
- Medical project: built a 3D shape reconstruction pipeline for custom orthosis design (pre-processing, segmentation, surface/mesh reconstruction, quality control) using OpenCV, scikit-image, Open3D; integrated outputs into clinical prototyping tools (CloudCompare/ParaView).
- Applied teacher–student distillation to compress YOLO models for embedded/edge constraints.
- Performed simulation/experiment coupling and validation using Radia magnetic-field simulations; ensured reproducibility and maintainability with Git (clean code, versioning, documented pipelines).

Oct 2020 - Sep 2024, ***Institut Jean Lamour (IJL), CNRS, University of Lorraine, Nancy, France***
PhD Candidate (AI for imaging physical systems) "Applications of artificial intelligence in plasma physics and machine vision".
- Designed and trained supervised deep-learning models for detection/segmentation of turbulent structures in ultra-fast imaging sequences; built end-to-end data pipelines (pre-processing, labeling strategy, evaluation metrics, model monitoring).
- Developed unsupervised / anomaly-detection approaches to characterize electric arc defects and bubble dynamics, including feature extraction and clustering/representation learning; leveraged Anomalib and custom Python tooling.
- Implemented computer vision and diagnostic calibration workflows (e.g., Calcam, Ax Vision) and maintained reproducible analysis codebases across experiments.

Oct 2021 - Nov 2021, ***Institue Plasma Physics (IPP), Prague***
Research Intern (Barrande Fellowship): Experimental/data-analysis work on COMPASS tokamak diagnostics; processed experimental datasets and contributed to interpretation/validation of analysis results.

Mars 2020 - Sep 2020, ***Centre de Recherche en Automatique de Nancy (CRAN)***
Graduation intership (Control / estimation): Developed observer-based methods for Li-ion battery management using equivalent electrical models; implemented estimation algorithms and validation workflows via Matlab/Simulink.

Sep 2020 - Feb 2020, ***Polytech Nancy***
School project: On-line estimation of physical parameters of dynamics systems.

Skills

Software
Matlab/Simulink | Ax Vision | Calcam | Proteus | Microsoft office | OpenOffice | CloudCompare | ParaView | Labview (Notion ++) | SolidWorks (Notion+)

Programming
Matlab / Simulink | Python | Latex | Arduino | C/C++ | R |Grafcet (Notion+) | Java (Notion++)

MLOps / DevOps
MLflow (Tracking, Projects, Model Registry) | CI/CD (GitHub Actions / GitLab CI) | Docker (Notion+)

High-Performance Computing (HPC)
CUDA | PyTorch | OpenCL | GPU kernels | Parallel computing | Deep learning on GPU

Jun 2019 - Jul 2019, *Laboratoire d'étude des Microstructures et de mécanique des matériaux (LEM3), Nancy*
Research intership Characterization of a piezoelectric material; creation of an AMF / Piezo system control interface using Labview.

Oct 2019 - May 2019, *Laboratoire d'étude des Microstructures et de mécanique des matériaux (LEM3), Nancy*
School project : Characterization of an energy recovery micro-device composed of a shape memory alloy / piezoelectric polymer.

Jul 2018, *Institut Jean Lamour, Nancy*
Research intership : Routing of an electronic card using Proteus software, selection of the appropriate components then making an optimized design of a routing scheme (PCB).

Oct 2017 - May 2018, *Polytech Nancy*
School project competition : Perception and electronics, Creation of an autonomous robot, with specific objectives according to the regulations of the CFR2018, La Roche-Sur-Yon, France.

Mar 2017, *Unité de Développement des Equipements Solaires (UDES)* / Dec 2016, *Schneider Electric, Algiers*
Beginner intership : Company discovery.

Publications:

Thesis manuscript:
Sarah Chouchene. Applications de l'intelligence artificielle à l'étude de la turbulence plasma en fusion nucléaire et aux plasmas d'arc en régime DC. Physique [physics]. Université de Lorraine, 2024. Français. ⟨NNT : 2024LORR0265⟩. ⟨tel-04961635⟩.

Articles:

- Lambert, M., Corentin, F. E. R. Y., Quirin, T., Vergne, C., Lemoigne, *S., Chouchene*, S., ... & Madec, M. (2025, June). Machine-Learning Algorithm Applied to Magnetic Localization. In 2025 23rd IEEE Interregional NEWCAS Conference (NEWCAS) (pp. 148-152). IEEE.

- *Chouchene, S.*, Brochard, F., Lemoine, N., Cavalier, J., Desecures, M., & Weinzettl, V. (2024). Mutual interactions between plasma filaments in a tokamak evidenced by fast imaging and machine learning. *Physical Review E*,109(4), 045201.

- *Chouchene, S.*, Brochard, F., Desecures, M., Lemoine, N., & Cavalier, J. (2024). Application of machine learning for detecting and tracking turbulent structures in plasma fusion devices using ultra fast imaging, *Sci Reports*, 14, 27965.

- Hellé, A., Hugon, R., Brochard, F., *Chouchene, S.*, Marcos, G., Moritz, J., & Schweitzer, P. (2024). Dynamics of the material ejection in a dipolar arc in continuous regime. *Journal of Physics D: Applied Physics*,57(42), 425201.

Conferences:

- Ouaaziz, M., klockenbring, D., Lam. J, Wemmert, C., Naegel, *Chouchene, S.*, Madec, M., Winfried, R., Guided Feature Distillation for YOLO11: Efficient Detection of Pseudomonas aeruginosa in High-Resolution Microscopy. Symposium on Intelligent Data Analysis (IDA 2026), Netherland.

- The 2nd International Annual Conference on Artificial Intelligence in Health (AiHealth). Reconstruction of 3D Shapes from Magnetic Data for Orthopedic Orthoses using Deep Learning. 10-11 November 2025, Basel (Switzerland).

- AI in Medicine & Healthcare : from algorithm to clinical application. 3D morphological reconstruction from magnetic data using deep learning, 5-6 November 2025. Strasbourg (France).

Communication and leadership
Organizer of different Projects and social events

Activities and interests

Association
Elite | OpenDay Polytech | Sigma Club

Technical interests
Multidisciplinary projects applying machine vision, deep learning, and 3D reconstruction (from research to deployment / clinical or industrial impact).

Hobbies
Traveling | Reading

Sport
Already practiced : Swimming | Kick Boxing | Aerobic gymnastics | Athletics

- *Chouchene, S.*, Brochard, F., Desécures, M., Lemoine, N., Cavalier, J., & Team, A. S. (Spokane 2022). Detection of Edge Plasma Turbulence Using Ultra Speed Camera and Artificial Intelligence. In *APS Division of Plasma Physics* Meeting Abstracts (Vol. 2022, pp. PO05-005) (USA).

- *LE STUDIUM* On-line Meeting on Artificial Intelligence for Plasma Science, 29-30 november 2022. Intitulée : Detection of Plasma turbulence in Plasma Fusion Devices Using Ultra-Fast Camera and Artificial Intelligence.

- 49th Conference on Plasma Physics (*EPS*) 03-07 july 2023, Bordeaux (France).

- 27th Joint EU-US Transport Task Force Meeting (*TTF*), 11-15 september 2023, Nancy (France).

Awards:

- Doctoral school C2MP Thesis Award 2024.